

Time Series — Solutions

Practice Exam — Week 2: ARMA Models and the PACF

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Problem 1 (5 points)

Statement.

(a) State the definition of white noise and write down its ACVS γ_τ and ACF ρ_τ . Explain in one sentence why “uncorrelated” is weaker than “independent”. (1.5 pts)

(b) Consider the MA(2) process

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}.$$

Compute the mean and the autocovariance γ_τ for all $\tau \in \mathbb{Z}$, and write down ρ_1 and ρ_2 . (2.5 pts)

(c) For which lags τ does the ACVS of an MA(q) vanish? State the general rule and explain how it is used to read off the order q from an estimated ACF. (1 pt)

Solution.

(a) $\{\varepsilon_t\}$ is *white noise* if it is a sequence of mean-zero, finite-variance, *uncorrelated* random variables: $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2 < \infty$, and $\text{Cov}(\varepsilon_s, \varepsilon_t) = 0$ for $s \neq t$. Its second-order structure is

$$\gamma_\tau = \begin{cases} \sigma^2, & \tau = 0, \\ 0, & \tau \neq 0, \end{cases} \quad \rho_\tau = \begin{cases} 1, & \tau = 0, \\ 0, & \tau \neq 0. \end{cases}$$

“Uncorrelated” only constrains the first two moments, whereas “independent” constrains the entire joint law; hence a white-noise process need not be i.i.d. (and need not be Gaussian).

(b) The constant mean is $\mathbb{E}[X_t] = \mathbb{E}[\varepsilon_t] - \theta_1 \mathbb{E}[\varepsilon_{t-1}] - \theta_2 \mathbb{E}[\varepsilon_{t-2}] = 0$. The filter taps are $(\psi_0, \psi_1, \psi_2) = (1, -\theta_1, -\theta_2)$. Using bilinearity and $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$, the ACVS is $\gamma_\tau = \sigma^2 \sum_j \psi_j \psi_{j+|\tau|}$ (only matching noise indices survive):

$$\begin{aligned} \gamma_0 &= \sigma^2 (1 + \theta_1^2 + \theta_2^2), \\ \gamma_{\pm 1} &= \sigma^2 [(1)(-\theta_1) + (-\theta_1)(-\theta_2)] = \sigma^2 \theta_1 (\theta_2 - 1), \\ \gamma_{\pm 2} &= \sigma^2 [(1)(-\theta_2)] = -\sigma^2 \theta_2, \\ \gamma_\tau &= 0 \quad (|\tau| \geq 3). \end{aligned}$$

Dividing by γ_0 ,

$$\rho_1 = \frac{\theta_1(\theta_2 - 1)}{1 + \theta_1^2 + \theta_2^2}, \quad \rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2}.$$

(c) For an MA(q) with taps $\psi_0, \dots, \psi_q$ one has $\gamma_\tau = \sigma^2 \sum_j \psi_j \psi_{j+|\tau|}$, and for $|\tau| > q$ no pair of taps overlaps (the filter has length $q + 1$), so $\gamma_\tau = 0$ for $|\tau| > q$. Thus the **ACVS of an MA(q) cuts off after lag q** : the largest lag at which the (sample) ACF is still significant — beyond the $\pm 1.96/\sqrt{n}$ band — identifies the order q .

Problem 2 (5 points)

Statement.

Consider the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$.

(a) State the condition on ϕ for the process to be causal, and explain it in terms of the root of the AR polynomial $\Phi(z) = 1 - \phi z$. (1 pt)

(b) Assuming $|\phi| < 1$, derive the causal MA(∞) representation $X_t = \sum_{k \geq 0} \psi_k \varepsilon_{t-k}$ by iterating the recursion. Give the ψ -weights and verify $\sum_k |\psi_k| < \infty$. (2 pts)

(c) Using the representation in (b), show that

$$\gamma_\tau = \frac{\sigma^2}{1 - \phi^2} \phi^{|\tau|}, \quad \rho_\tau = \phi^{|\tau|}.$$

Why does this ACF tail off rather than cut off, and what does that say about identifying p from the ACF? (2 pts)

Solution.

(a) The process is *causal* iff $|\phi| < 1$. The AR polynomial $\Phi(z) = 1 - \phi z$ has its single root at $z = 1/\phi$, and the causality condition “root outside the unit circle” reads $|1/\phi| > 1 \iff |\phi| < 1$.

(b) Iterating $X_t = \phi X_{t-1} + \varepsilon_t$, with $X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$, etc.:

$$\begin{aligned} X_t &= \varepsilon_t + \phi X_{t-1} = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 X_{t-2} \\ &= \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \cdots + \phi^{n-1} \varepsilon_{t-n+1} + \phi^n X_{t-n}. \end{aligned}$$

Since $|\phi| < 1$ and $\text{Var}(X_{t-n}) = \gamma_0$ is constant, the remainder satisfies $\mathbb{E}[(\phi^n X_{t-n})^2] = \phi^{2n} \gamma_0 \rightarrow 0$, so it vanishes in mean square and

$$X_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}, \quad \psi_k = \phi^k.$$

The weights are absolutely summable: $\sum_{k \geq 0} |\psi_k| = \sum_{k \geq 0} |\phi|^k = \frac{1}{1 - |\phi|} < \infty$, confirming a valid causal MA(∞) representation.

(c) With $\mathbb{E}[X_t] = 0$ and $\text{Cov}(\varepsilon_{t-k}, \varepsilon_{t-\tau-l}) = \sigma^2 \mathbf{1}_{\{t-k=t-\tau-l\}}$, take $\tau \geq 0$:

$$\begin{aligned} \gamma_\tau &= \text{Cov}\left(\sum_{k \geq 0} \phi^k \varepsilon_{t-k}, \sum_{l \geq 0} \phi^l \varepsilon_{t-\tau-l}\right) = \sigma^2 \sum_{l \geq 0} \phi^{l+\tau} \phi^l \\ &= \sigma^2 \phi^\tau \sum_{l \geq 0} \phi^{2l} = \frac{\sigma^2 \phi^\tau}{1 - \phi^2}, \end{aligned}$$

where the surviving terms are those with $k = l + \tau$, and the geometric series converges since $\phi^2 < 1$. By symmetry $\gamma_\tau = \frac{\sigma^2}{1 - \phi^2} \phi^{|\tau|}$, and $\rho_\tau = \gamma_\tau / \gamma_0 = \phi^{|\tau|}$.

The ACF $\phi^{|\tau|}$ decays geometrically but is never exactly 0 for $|\phi| \in (0, 1)$, so it *tails off* rather than cuts off. Consequently the order p of an AR cannot be read from the ACF; it is the **PACF** that cuts off at lag p (here $p = 1$).

Problem 3 (5 points)

Statement.

(a) Consider the AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ with $\phi_1 = \frac{1}{2}$, $\phi_2 = -\frac{1}{4}$. Using the causality conditions $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$, verify that the process is causal. (1 pt)

(b) From the Yule–Walker equations $\rho_\tau = \phi_1 \rho_{\tau-1} + \phi_2 \rho_{\tau-2}$ (with $\rho_0 = 1$, $\rho_{-1} = \rho_1$), compute ρ_1, ρ_2, ρ_3 . (2 pts)

(c) Using the Durbin–Levinson recursion

$$\alpha_{1,1} = \rho_1, \quad \alpha_{\tau,\tau} = \frac{\rho_\tau - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_j}, \quad \alpha_{\tau,j} = \alpha_{\tau-1,j} - \alpha_{\tau,\tau} \alpha_{\tau-1,\tau-j},$$

compute $\alpha_1, \alpha_2, \alpha_3$ from the values in (b). Read off the implied model order and state the ACF/PACF identification rule it illustrates. (2 pts)

Solution.

(a) With $\phi_1 = \frac{1}{2}$, $\phi_2 = -\frac{1}{4}$:

$$\phi_1 + \phi_2 = \frac{1}{4} < 1, \quad \phi_2 - \phi_1 = -\frac{3}{4} < 1, \quad |\phi_2| = \frac{1}{4} < 1.$$

All three causality conditions hold, so the AR(2) is causal (both roots of $\Phi(z) = 1 - \frac{1}{2}z + \frac{1}{4}z^2$ lie outside the unit circle).

(b) The Yule-Walker recursion $\rho_\tau = \phi_1\rho_{\tau-1} + \phi_2\rho_{\tau-2}$. At $\tau = 1$, using $\rho_{-1} = \rho_1$, we get $\rho_1 = \phi_1\rho_0 + \phi_2\rho_1 = \phi_1 + \phi_2\rho_1$, so

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} = \frac{1/2}{1 + 1/4} = \frac{1/2}{5/4} = \frac{2}{5}.$$

Then

$$\begin{aligned} \rho_2 &= \phi_1\rho_1 + \phi_2 = \frac{1}{2} \cdot \frac{2}{5} - \frac{1}{4} = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}, \\ \rho_3 &= \phi_1\rho_2 + \phi_2\rho_1 = \frac{1}{2} \cdot \left(-\frac{1}{20}\right) - \frac{1}{4} \cdot \frac{2}{5} = -\frac{1}{40} - \frac{1}{10} = -\frac{1}{40} - \frac{4}{40} = -\frac{5}{40} = -\frac{1}{8}. \end{aligned}$$

(c) Durbin-Levinson with $\rho_1 = \frac{2}{5}$, $\rho_2 = -\frac{1}{20}$, $\rho_3 = -\frac{1}{8}$.

Lag 1: $\alpha_{1,1} = \rho_1 = \frac{2}{5}$.

Lag 2:

$$\alpha_{2,2} = \frac{\rho_2 - \alpha_{1,1}\rho_1}{1 - \alpha_{1,1}\rho_1} = \frac{-\frac{1}{20} - \frac{2}{5} \cdot \frac{2}{5}}{1 - \frac{2}{5} \cdot \frac{2}{5}} = \frac{-\frac{1}{20} - \frac{4}{25}}{1 - \frac{4}{25}} = \frac{-\frac{21}{100}}{\frac{84}{100}} = -\frac{21}{84} = -\frac{1}{4}.$$

Update: $\alpha_{2,1} = \alpha_{1,1} - \alpha_{2,2}\alpha_{1,1} = \frac{2}{5} - \left(-\frac{1}{4}\right)\frac{2}{5} = \frac{2}{5} + \frac{1}{10} = \frac{1}{2}$.

Lag 3:

$$\alpha_{3,3} = \frac{\rho_3 - \alpha_{2,1}\rho_2 - \alpha_{2,2}\rho_1}{1 - \alpha_{2,1}\rho_1 - \alpha_{2,2}\rho_2}.$$

The numerator is

$$-\frac{1}{8} - \frac{1}{2} \cdot \left(-\frac{1}{20}\right) - \left(-\frac{1}{4}\right) \cdot \frac{2}{5} = -\frac{1}{8} + \frac{1}{40} + \frac{1}{10} = -\frac{5}{40} + \frac{1}{40} + \frac{4}{40} = 0,$$

so $\alpha_{3,3} = 0$ (the denominator $1 - \frac{1}{2} \cdot \frac{2}{5} - (-\frac{1}{4})(-\frac{1}{20}) = 1 - \frac{1}{5} - \frac{1}{80} > 0$ is nonzero). Hence

$$\alpha_1 = \frac{2}{5}, \quad \alpha_2 = -\frac{1}{4}, \quad \alpha_3 = 0.$$

The PACF cuts off after lag 2 while the ACF in (b) tails off: this is the **AR(p) signature with $p = 2$** . Indeed the diagonal coefficients recover the AR coefficients, $\alpha_{2,1} = \frac{1}{2} = \phi_1$ and $\alpha_{2,2} = -\frac{1}{4} = \phi_2$. (Identification rule: PACF cuts after lag p , ACF tails off $\Rightarrow$ AR(p).)

Problem 4 (5 points)

Statement.

Consider the ARMA(1,1) process $X_t = \phi X_{t-1} + \varepsilon_t + \vartheta \varepsilon_{t-1}$ with $\phi = \frac{1}{2}$, $\vartheta = \frac{1}{4}$.

(a) Write the model as $\Phi(B)X_t = \Theta(B)\varepsilon_t$ and check causality and invertibility from the roots of $\Phi(z)$ and $\Theta(z)$. (1.5 pts)

(b) Derive the ψ -weights of the causal MA(∞) representation $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ by matching coefficients in $\Phi(B)\psi(B) = \Theta(B)$. Give ψ_0, ψ_1, ψ_2 and the general formula. (2 pts)

(c) (Week 1 revision.) Recall that a candidate sequence $\{\gamma_\tau\}$ is a valid ACVS only if it is positive semi-definite. Decide whether

$$\gamma_0 = 1, \quad \gamma_{\pm 1} = 0.7, \quad \gamma_\tau = 0 \quad (|\tau| \geq 2)$$

is a valid ACVS of some stationary process, by computing the spectral density $\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau}$ and checking its sign. (1.5 pts)

Solution.

(a) In backshift form $X_t - \phi X_{t-1} = \varepsilon_t + \vartheta \varepsilon_{t-1}$ reads $\Phi(B)X_t = \Theta(B)\varepsilon_t$ with

$$\Phi(z) = 1 - \phi z = 1 - \frac{1}{2}z, \quad \Theta(z) = 1 + \vartheta z = 1 + \frac{1}{4}z.$$

$\Phi(z)$ has root $z = 1/\phi = 2$, with $|2| > 1 \Rightarrow$ **causal**. $\Theta(z)$ has root $z = -1/\vartheta = -4$, with $|-4| > 1 \Rightarrow$ **invertible**.

(b) Write $\psi(z) = \sum_{j \geq 0} \psi_j z^j$ and match coefficients in $\Phi(z)\psi(z) = \Theta(z)$, i.e.

$$(1 - \phi z) \sum_{j \geq 0} \psi_j z^j = 1 + \vartheta z.$$

Coefficient of z^0 : $\psi_0 = 1$. Coefficient of z^1 : $\psi_1 - \phi\psi_0 = \vartheta$, so $\psi_1 = \phi + \vartheta$. Coefficient of z^j for $j \geq 2$: $\psi_j - \phi\psi_{j-1} = 0$, so $\psi_j = \phi\psi_{j-1}$. Solving the recursion gives the general formula

$$\psi_0 = 1, \quad \psi_j = (\phi + \vartheta) \phi^{j-1} \quad (j \geq 1).$$

With $\phi = \frac{1}{2}$, $\vartheta = \frac{1}{4}$ we have $\phi + \vartheta = \frac{3}{4}$, hence

$$\psi_0 = 1, \quad \psi_1 = \frac{3}{4}, \quad \psi_2 = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}.$$

(The weights decay like ϕ^j , so $\sum_j |\psi_j| = 1 + \frac{3}{4} \sum_{j \geq 0} (\frac{1}{2})^j = 1 + \frac{3}{4} \cdot 2 = \frac{5}{2} < \infty$, confirming causality.)

(c) **No**, the sequence is not a valid ACVS. A valid ACVS must be positive semi-definite, equivalently have a non-negative spectral density. Here, using $\gamma_{-1} = \gamma_1$,

$$\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau} = \gamma_0 + 2\gamma_1 \cos(2\pi f) = 1 + 1.4 \cos(2\pi f).$$

At $f = \frac{1}{2}$ this equals $1 + 1.4 \cos(\pi) = 1 - 1.4 = -0.6 < 0$. Since the spectral density goes negative, the sequence fails positive semi-definiteness and *cannot* be the ACVS of any stationary process. (Equivalently $\rho_1 = 0.7 > \frac{1}{2}$, which is impossible for an MA(1)-type two-tap correlation.)